

Compiled with problems:

ERROR in ./docs/semester-1/gqur-300/unit-03/topic-01.mdx

Module build failed (from ./node_modules/@docusaurus/mdx-loader/lib/index.js):

Error: MDX compilation failed for file

["/home/a2ahs/mega_book_for_B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/gqur-300/unit-03/topic-01.mdx"](#)

Cause: Unexpected character `=` (U+003D) before name, expected a character that can start a name, such as a letter, `\$`, or `\_`

Details:

```
{
  "column": 36,
  "file": "",
  "message": "Unexpected character `=` (U+003D) before name,
expected a character that can start a name, such as a
letter, `$`, or `_`",
  "line": 141,
  "name": "141:36",
  "place": {
    "_bufferIndex": 33,
    "_index": 0,
    "line": 141,
    "column": 36,
    "offset": 7642
  },
  "reason": "Unexpected character `=` (U+003D) before name,
expected a character that can start a name, such as a
letter, `$`, or `_`",
  "ruleId": "unexpected-character",
  "source": "micromark-extension-mdx-jsx",
  "url": "https://github.com/micromark/micromark-extension-
mdx-jsx#unexpected-character-at-expected-expect"
}
```

at compileToJSX

reasoning the cook already owns, names it, and connects it to the percentage table that will follow, so you can use it deliberately and teach it explicitly.

Explanation

Ratios and proportions

A ratio compares two quantities of the same kind: 6 spoons of sugar to 4 spoons of tea leaves, 40 girls to 20 boys in a school. A proportion is a statement that two ratios are equal: 6 to 4 equals 18 to 12, because both are 3 to 2. The cross-products check makes a proportion verifiable without trusting the eye: $6 \times 12 = 72$ and $4 \times 18 = 72$, so the proportion holds.

Worked example (scaling up). The recipe serves 20 people with 6 spoons of sugar. For 60 people: the crowd is 3 times as large, so every ingredient is 3 times as large: 18 spoons. For 50 people: the crowd is 2.5 times as large, so 15 spoons. The reasoning move is always the same: find the multiplier between the situations and apply it to every quantity.

Worked example (finding the multiplier). A school has 40 girls and 20 boys. The ratio of girls to boys is 40 to 20, which simplifies to 2 to 1: two girls for every boy. If the next school year admits 30 more pupils in the same ratio, expect 20 girls and 10 boys.

A common misconception to correct. Trainees often add instead of multiplying when scaling: "20 more people, so 6 more spoons of sugar." Adding the same amount keeps the difference constant, but recipes need the ratio constant. The test is to ask what must stay the same: the taste needs the ratio, so multiply, never add.

Percentages

Situation	What it asks	Worked rupee example
Percentage of a quantity	take a named share of the whole	15 percent of a 2,000 rupee fee is 300 rupees
Percentage increase	grow the original by a named share	a 1,200 rupee bill rising by 10 percent becomes 1,320
Percentage decrease	shrink the original by a named share	a 500 rupee shirt at 20 percent off costs 400 rupees
Reverse percentage	find the original before a known change	after a 10 percent rise the fee is 2,200, so it was 2,000

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A percentage is a ratio expressed out of 100: "per cent" literally means "by the hundred". Percentages matter because so many daily quantities are quoted in them: fee rises, discounts, exam marks, battery levels, tax. The table above sorts the four situations you will meet, each with a rupee example. The fourth row deserves special care: finding the original after a rise is the classic trap, because the rise acts on the original, not on the final figure.

Worked example (marks). A pupil scores 36 out of 45 in a math test. As a percentage: 36 divided by 45 is 0.8, which is 80 percent. The reasoning: the fraction 36/45 tells what share of the marks was earned; multiplying by 100 renames that share per hundred.

Worked example (discount). A bookshop offers 25 percent off a 640-rupee atlas. The discount is $25/100 \times 640 = 160$ rupees, so the sale price is $640 - 160 = 480$ rupees. A faster route: paying 75 percent of the price is $0.75 \times 640 = 480$; both routes must agree, and checking that they do is the look-back habit from Unit 1.

Activity: The marksheet percentage relay

Work in groups of four. Your tutor gives each group five anonymised test results in different totals (for example 36/45, 42/60, 55/75, 18/25, 27/30). Each member converts two results to percentages by hand (no calculator), then the group ranks all five pupils from best to worst percentage. The debrief question: which raw scores looked biggest but were not best, and why the total matters. About 20 minutes.

Check your understanding

1. Write the ratio 40 girls to 20 boys in its simplest form, and state what it means in one sentence. (*Understand*)
2. A recipe for 12 people uses 4 eggs. How many eggs for 30 people? Show the multiplier you used. (*Apply*)
3. Compute 15 percent of a 2,000-rupee fee, and state the new fee after a 15 percent rise. (*Apply*)
4. After a 10 percent rise, a school's monthly electricity bill is 17,600 rupees. What was the bill before the rise? Show your reasoning. (*Apply*)
5. A friend scales a recipe by adding: "20 more people, so add 6 more spoons of sugar." Explain in two sentences why this breaks the recipe and what to do instead. (*Analyze*)

Summary

Ratio compares, proportion states that two ratios are equal, and percentage is a ratio out of 100; they are one idea in three notations. Scaling multiplies every ingredient by the same multiplier - it never adds. The four percentage situations (of, increase, decrease, reverse) each need their own setup, and the reverse case is the classic trap because the rise acts on the original.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can simplify a ratio and explain what it means in one sentence.
- ☐ I can scale a recipe or a plan by finding and applying the multiplier.
- ☐ I can handle all four percentage situations with rupee examples, including the reverse case.

Try this at your practicum school

Take one class's latest test (anonymised, with permission). Convert any five pupils' raw scores to percentages. Then ask the teacher one question: does the school report raw scores or percentages to parents, and why? Note the answer with one example.

Summative task

A stationery supplier offers two deals on notebooks costing 80 rupees each: Deal A gives "20 percent off", Deal B gives "5 notebooks free with every 20 bought". The school needs 100 notebooks. In a short written answer: (a) compute the total cost under each deal, (b) state which deal is cheaper and by how much, and (c) identify one circumstance (a different quantity, budget timing, or storage) under which the school might still prefer the other deal, with a reason.

Mini-rubric (each criterion 0-2 marks):

- *Both totals correct:* Deal A costs 6,400 rupees ($80 \times 100 \times 0.8$); Deal B costs 6,400 rupees (pay 80, receive 100) - or the equivalent correct computation shown.
- *Comparison stated:* the cheaper deal named with the difference (here they tie, and noticing the tie earns the mark).
- *Circumstance analysed:* a real non-price factor is identified and argued, not merely named. **(Analyze)**

Further reading

- Marecek, L., Anthony-Smith, M., & Mathis, M. H. (2020). *Prealgebra 2e* (Ch. 6 Percents; Ch. 5, Section 5.6 Ratios and Rate). OpenStax.
<https://openstax.org/details/books/prealgebra-2e>
- National Curriculum Council, Ministry of Federal Education and Professional Training, Government of Pakistan. *Mathematics: Suggested Guidelines (Grades 1-8)*.
<https://ncc.gov.pk/Detail/ZjgzYzg2MmMtZDc0Zi00NjEzLTk5ZmYtZGJiNjc5ODljOGUx>

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

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